

From Concept Identification to Concept Removal

Benchmarking CAV-Based Methods for Debiasing Multimodal Models

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Abstract

Deep neural networks encode high-level semantic concepts in their latent representations, which can improve predictive performance, but may also introduce undesirable biases. While Concept Activation Vectors (CAVs) have been widely used for interpretability and model steering, their use for direct removing target concepts from internal representations remains largely unexplored.

We investigate concept removal as a debiasing strategy for multimodal models and introduce three novel hidden-layer interventions based on alternative approaches to CAV estimation and concept ablation. To systematically evaluate these methods and compare them to known procedures, we propose a unified benchmark that measures both concept suppression and preservation of model utility.

Using the CLIP ViT-L/14 vision-language model and the CelebA dataset, we demonstrate the effectiveness of the proposed approaches in reducing the detectability of target concepts while maintaining performance on non-target attributes. Our results provide new insights into the trade-off between bias mitigation and task preservation in multimodal neural networks, demonstrating a 94% reduction in median concept detectability at the cost of only a marginal drop in test set balanced accuracy (from 61.3% to 60.9%), while alternatively showcasing the capacity to enhance primary model performance from 61.3% to 61.8% median accuracy via the proposed debiasing framework.

1. Introduction and Related Work

Deep neural network architectures often encode high-level semantic concepts in their hidden representations. While existence of such concepts can improve predictive performance, it may also introduce undesirable biases when models rely on sensitive attributes such as gender, ethnicity or age. Understanding and controlling these latent concepts is therefore a critical challenge for developing fair and trustworthy machine learning systems.

Concept Activation Vectors (CAVs) (Kim et al., 2018) provide a powerful framework for identifying and manipulating human-interpretable concepts within neural representations. By representing a concept as a direction in the latent space, CAVs enable both the analysis of concept influence on model predictions and targeted interventions within internal representations.

To date, activation and steering vectors have primarily been leveraged to guide model behavior (Bayat et al., 2025; Li et al., 2023; Erogullari et al., 2025) or to provide post-hoc explainability regarding internal operations (Kim et al., 2018). However, to the best of our knowledge, employing CAVs specifically to eradicate the latent representation of a target concept from a model's internal processing remains largely unexplored.

Our contributions are as follows: **1)** We introduce a unified benchmark for concept removal techniques, evaluating both overall model performance and concept erasure efficacy; **2)** we propose three novel CAV-based debiasing procedures and collate them with an existing CAV-based method; **3)** we demonstrate that our debiasing methods can either significantly impair concept detection (reducing median accuracy by 94% across 5 debiased concepts) while slightly ($61.3\% \rightarrow 60.9\%$) deteriorate median balanced accuracy on 35 other concepts) or enhance overall performance via hidden-layer interventions (improving median test set accuracy from 61.3% to 61.8%). All experiments are conducted using the multimodal CLIP ViT-L/14 model (Radford et al., 2021) on datasets derived from CelebA (Liu et al., 2015).

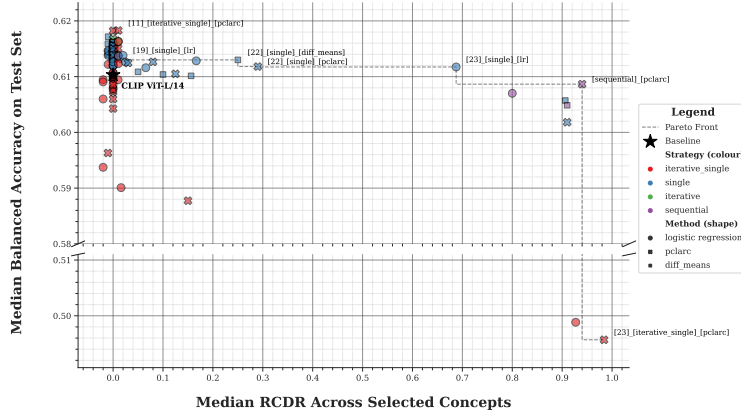


Figure 1: Performance comparison of debiasing procedures on CLIP ViT-L/14. The x-axis shows the median percentage of the erased concept (RCDR, see Eq. 1), while the y-axis represents the median balanced accuracy (BA) on test set derived from CelebA. Marker shapes and colors denote the CAV fixing method and debiasing strategy, respectively. The original model (baseline) is indicated by a black star. Our proposed methods can erase up to 94% of the concept slightly degrading test performance ($61.3\% \rightarrow 60.9\%$ BA), and in some configurations even improve overall accuracy. Compared to the known **single** debiasing method (blue circles/squares), our approaches achieve superior outcomes at both extremes (maximizing either concept erasure or performance boost). Finally, we note that complete concept erasure invariably causes severe model degradation, reducing accuracy to near-random levels (50%).

2. Methods

Fixing the CAV. We consider two methods for estimating Concept Activation Vectors (CAVs). The first, adopted from (Bayat et al., 2025), defines the CAV as the difference between the mean representations of positive and negative samples, denoted as CAV_{df} . The second, inspired by (Kim et al., 2018), trains a logistic classifier, namely logistic regression and uses its weights parameter vector β as the CAV. Since β is orthogonal to the learned decision hyperplane, it captures the direction most predictive of the target concept; we denote this vector by CAV_{lr} .

Debiasing with CAVs. After estimating the CAV, we use it to remove concept-related information from model representations. To this end, we evaluate two debiasing approaches. The first projects representations onto the subspace orthogonal to the CAV, thereby elim-

inating the component associated with the target concept. The second applies P-ClArC (Anders et al., 2022), which shifts positive samples towards the negative class along the CAV direction. Combining the two CAV estimation methods with the debiasing methods yields three experimental configurations: **1) diff_means**, which uses orthogonal projection with CAV_{df} ; **2) lr**, which uses orthogonal projection with CAV_{lr} ; and **3) pclarc**, which applies P-ClArC using CAV_{df} .

Debiasing strategy. Once the CAV and debiasing method are selected, it is necessary to determine at which layer(s) the intervention is applied. To this end, we consider four debiasing strategies, all implemented through hooks that modify hidden activations before they are passed to the subsequent layer. Specifically, already known in literature **single** method (Anders et al., 2022) applies debiasing at a single fixed hidden layer, whereas **sequential** method applies it independently at every hidden layer. The **iterative_single** strategy performs debiasing repeatedly for n iterations at a selected layer, using the output of each iteration as the input to the next, while **iterative_sequential** extends this procedure to all hidden layers. Since the **diff_means** method removes the entire component aligned with the CAV in a single projection step, iterative strategies are evaluated only with the **lr** and **pclarc** methods. We refer to each unique combination of a strategy and a debiasing method as a *procedure*.

3. Experiments

Concept selection and dataset preparation. Due to computational constraints, we restrict our analysis to five CelebA concepts exhibiting varying levels of baseline detectability with CLIP ViT-L/14: *male* (98.2%), *eyeglasses* (90.1%), *wearing_hat* (89.6%), *bald* (74.7%), and *chubby* (61.2%) (see Figure A.2). For evaluation, we construct a balanced test set of 256 images (1:1 positive-to-negative ratio) for each concept, regardless of whether the concept is targeted for debiasing. In contrast, balanced training subsets comprising 1,792 images are prepared only for concepts selected for removal and are used to compute the corresponding CAVs. Importantly, overall model utility is assessed on the remaining 35 CelebA attributes, which are never targeted by any debiasing intervention.

Evaluation Protocol. Following zero-shot classification (Radford et al., 2021), text representations are generated by mean-pooling and ℓ_2 -normalizing CLIP ViT-L/14 text embeddings from 3–4 custom templates (see Table A.2). At inference, we compute the cosine similarity between these text vectors and image embeddings from the original or debiased vision encoder. Scores are temperature-scaled ($\tau = 1000$), softmax-normalized and binarized at a 0.5 threshold. To quantify erasure efficacy, we introduce the **Removed Concept Detection Ratio (RCDR)**:

$$\text{RCDR} = \frac{y - \tilde{y}}{y - y_{rand}} \quad (1)$$

where y and $\tilde{y}$ denote the baseline and debiased model accuracies on the target concept test split respectively, and $y_{rand} = 0.5$ represents performance corresponding to random decisions. RCDR measures the fraction of discriminative ability mitigated by the intervention. Values outside $[0, 1]$ indicate either unintended detection improvement (< 0) or over-cleansing below chance level (> 1). Downstream utility is evaluated on the test set via balanced accuracy across the remaining 35 unmanipulated CelebA attributes.

Results As illustrated on Figure 1 and detailed in Table 1 our evaluation reveals a strict trade-off between concept removal efficiency (RCDR) and the preservation of models’s performance on the test set. We identify `sequential` strategy with `pclarc` debiasing method as the most effective approach for balancing these objectives with median RCDR equal to 0.94, whereas median BA is worse than baseline by just 0.653% (from 0.613 to 0.609). Moreover, we observe that procedures including interventions on the last layer successfully mitigate target concepts while maintaining model’s performance evaluated on the test set. Conversely, shifting the debiasing intervention e.g to layer 11 completely fails to remove completely fails to remove the target concept, yielding near-zero RCDR values, despite slightly improving downstream accuracy. Furthermore, aggressive attempts to entirely eradicate the concept such as the `iterative_single` with `lr` in the last layer result in severe degradation of the model’s overall discriminative capabilities. Lastly, as shown in Table 1 and Figure A.3, the most effective way to erase concept from model is to perform debiasing at the last layer.

Table 1: Concept erasure results on CLIP ViT-L/14. Procedure labels denote strategy, layer, and debiasing method. Metrics include erased concept detection accuracy (RCDR, Eq. 1) and balanced accuracy (BA) on 35 untouched CelebA attributes. We report that erasure is highly concept-dependent, with no single configuration consistently excelling across all concepts. Moreover, unlike late-layer interventions, early-layer debiasing achieves near-zero RCDR while preserving or improving baseline utility (original CLIP ViT-L/14 BA = 0.613). Comprehensive results for all conducted experiments are shown in Table A.3.

Debiased concept Procedure	bald		chubby		eyeglasses		male		wearing_hat		Median	
	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR
single at 01, lr	<u>0.617</u>	0.000	0.612	-0.025	0.608	-0.010	<u>0.614</u>	0.000	0.618	-0.010	<u>0.614</u>	-0.010
single at 10, pclarc	0.615	0.000	<u>0.609</u>	0.000	0.606	-0.010	<u>0.614</u>	0.000	0.616	-0.010	<u>0.614</u>	0.000
iterative_single at 11, pclarc	0.619	0.033	0.601	0.025	0.604	-0.050	0.618	0.000	0.623	0.010	0.618	0.010
single at 22, diff_means	0.613	0.393	0.601	0.225	0.613	0.250	0.610	0.056	<u>0.620</u>	0.323	0.613	0.250
single at 22, lr	0.613	0.279	0.608	0.025	0.609	0.320	<u>0.614</u>	0.008	0.617	0.167	0.613	0.167
single at 22, pclarc	0.612	0.393	0.607	0.200	<u>0.612</u>	0.290	0.609	0.304	0.613	0.219	0.612	0.290
single at 23, diff_means	0.606	<u>0.951</u>	0.593	0.500	0.606	0.900	0.598	0.968	0.617	0.906	0.606	0.906
single at 23, lr	0.610	0.639	<u>0.609</u>	0.025	<u>0.612</u>	0.710	<u>0.614</u>	0.760	0.618	0.688	0.612	0.688
single at 23, pclarc	0.602	0.967	0.596	0.425	0.607	<u>0.910</u>	0.590	<u>0.992</u>	0.613	0.562	0.602	<u>0.910</u>
sequential, diff_means	0.606	0.934	0.590	<u>0.475</u>	0.605	<u>0.910</u>	0.595	0.912	0.614	0.698	0.605	<u>0.910</u>
sequential, lr	0.607	0.820	0.604	0.250	0.607	0.870	0.606	0.800	0.616	<u>0.729</u>	0.607	0.800
sequential, pclarc	0.610	<u>0.951</u>	0.598	0.375	0.609	0.940	0.591	1.000	0.609	0.604	0.609	0.940

4. Discussion

Our findings (Figure 1, Table 1) demonstrate that the proposed debiasing procedures based on CAVs effectively debias models by achieving high Removed Concept Detection Ratio (RCDR) scores while preserving overall utility, although this efficacy remains highly concept-dependent. For instance, while the *bald* attribute is nearly eradicated, the *chubby* concept exhibits significant resistance. Also, it is impossible to achieve both full concept removal and preserving model overall performance in other fields at the same time. Consequently, future research should expand more advanced evaluations across diverse model architectures and broader attribute sets to elucidate the underlying causes of this variable susceptibility to debiasing.

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5. Appendix

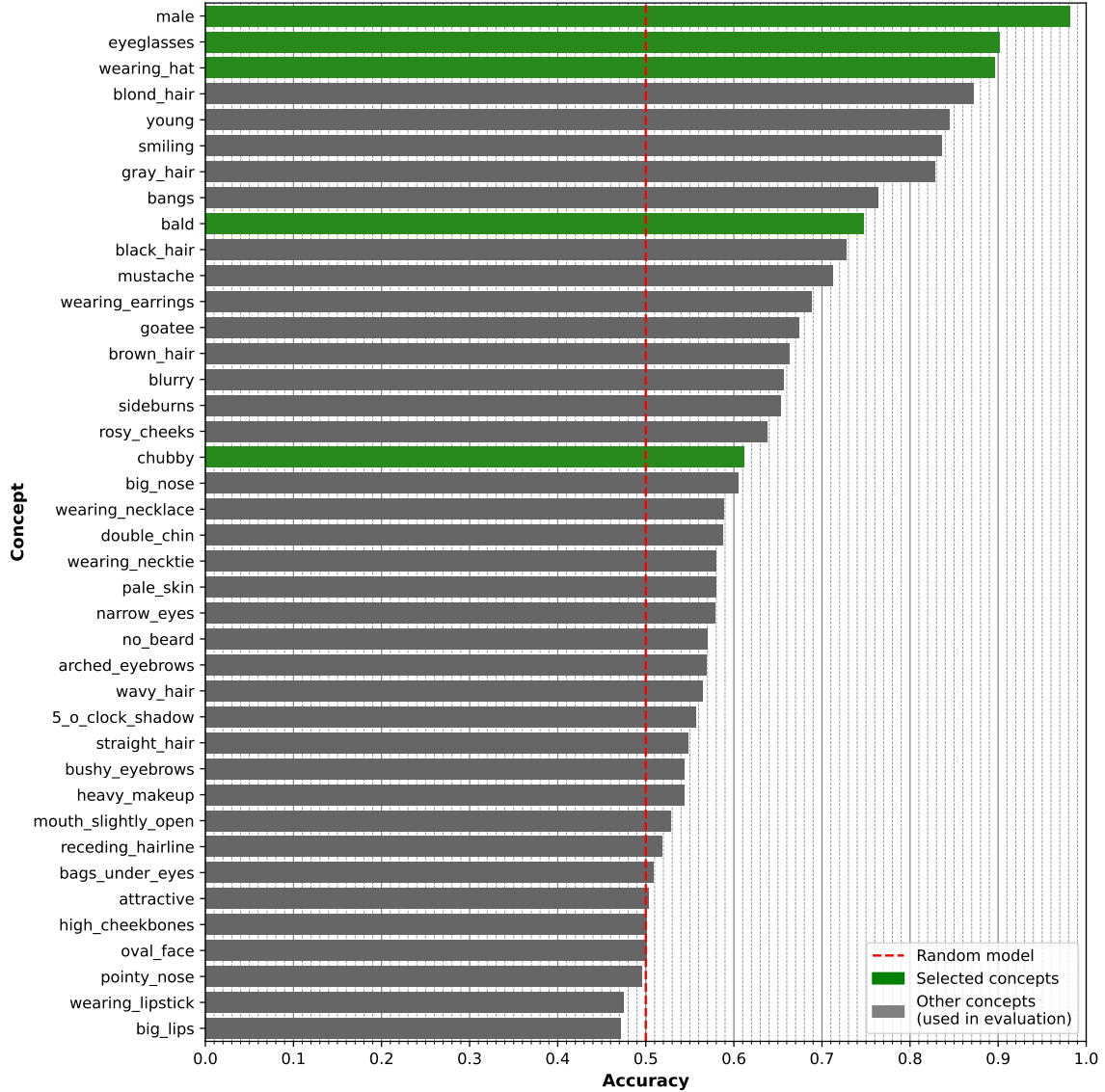


Figure 2: Concept detection performance of the zero-shot evaluation method, utilizing prompt embeddings specified in Table 2 with the CLIP ViT-L/14 model across all CelebA dataset attributes. Each concept is evaluated on a balanced subset of 256 images (128 positive and 128 negative samples). The balanced accuracy varies substantially across concepts, ranging from highly discriminative features (e.g., *male*: 0.982, *eyeglasses*: 0.901) to those performing near chance level (e.g., *big_lips*: 0.500). Green bars denote target attributes selected for evaluating debiasing interventions, whereas gray bars represent the remaining evaluation subset used to assess model performance on unbiased concepts.

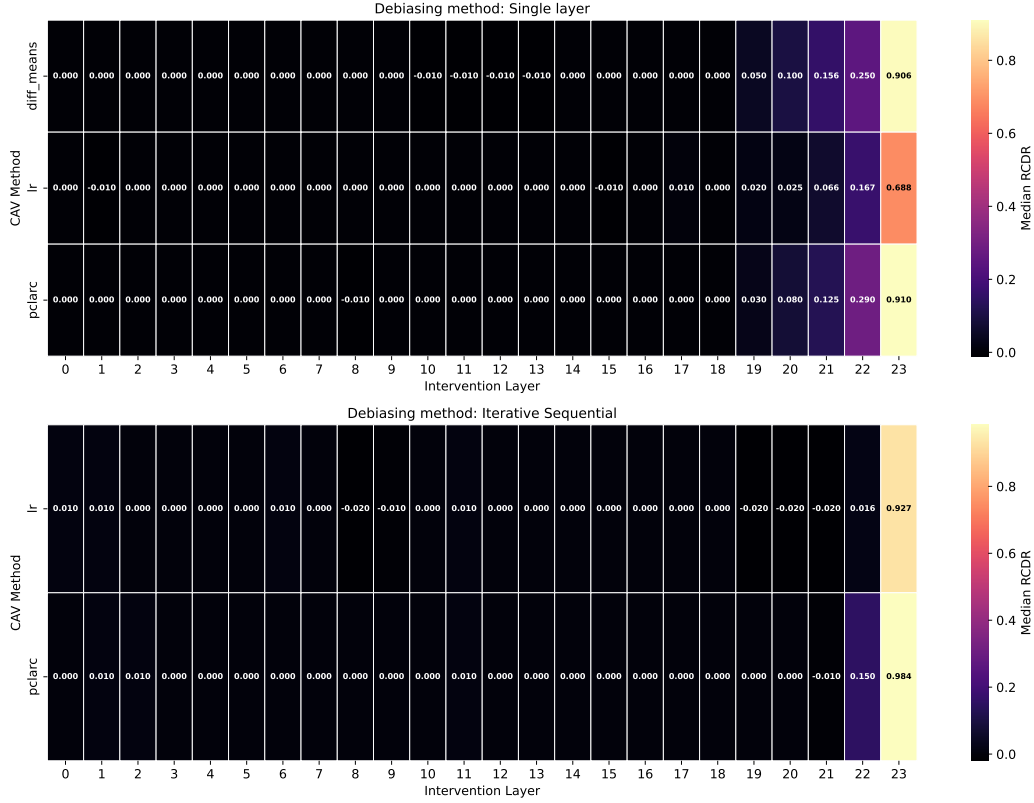


Figure 3: Median removed concept detection (RCDR, see Eq. 1) in zero-shot classification of debiased concepts using single-layer debiasing methods (**single** and **iterative-single**). In both cases actual changes in concept removal effectiveness are visible only when these methods are applied to the last hidden layers of the model. Applying the methods to earlier layers has no observable effect on the model’s detection of the concept during zero-shot classification. In the case of the **iterative-single** method, no experiments were conducted for the **diff_means** CAV, which is why no results are available for it.

Table 2: Comprehensive list of custom text templates utilized for prompt ensembling in the zero-shot classification framework. For each of the 40 CelebA attributes, we specify 3–4 positive prompts (indicating concept presence) and corresponding negative prompts (indicating concept absence). Consistent with our evaluation protocol, the text embeddings of these templates are mean-pooled and ℓ_2 -normalized to construct a single, robust reference vector for each class.

Concept	Positive prompts	Negative Prompts
5 o'clock shadow	• a photo of a person with a five o'clock shadow • a photo of a person with light stubble on their face • a photo of a man with short facial stubble • a photo of a person with a day's worth of beard growth	• a photo of a person with a clean-shaven face • a photo of a person with no facial stubble • a photo of a person with smooth skin and no beard growth
arched eyebrows	• a photo of a person with arched eyebrows • a photo of a person with highly curved eyebrows • a photo of a person with elegantly arched brows	• a photo of a person with straight eyebrows • a photo of a person with flat, unarched eyebrows • a photo of a person with horizontal brows
attractive	• a photo of an attractive person • a photo of a good-looking person • a photo of a beautiful person • a photo of a handsome or pretty person	• a photo of a plain-looking person • a photo of an average-looking person • a photo of an unattractive person
bags under eyes	• a photo of a person with bags under their eyes • a photo of a person with puffy under-eye areas • a photo of a person with dark circles and swelling under the eyes • a photo of a tired-looking person with under-eye bags	• a photo of a person with no bags under their eyes • a photo of a person with smooth skin under the eyes • a photo of a person with fresh, rested-looking eyes
bald	• a photo of a bald person • a photo of a person with no hair on their head • a photo of a person with a shaved head • a photo of a person who is completely bald	• a photo of a person with hair on their head • a photo of a person with a full head of hair • a photo of a person who is not bald
bangs	• a photo of a person with bangs • a photo of a person with hair covering their forehead • a photo of a person with a fringe hairstyle • a photo of a person with bangs across their forehead	• a photo of a person without bangs • a photo of a person with their forehead fully visible • a photo of a person with hair swept back from the forehead
big lips	• a photo of a person with big lips • a photo of a person with full, prominent lips • a photo of a person with large, voluminous lips	• a photo of a person with thin lips • a photo of a person with small lips • a photo of a person with narrow lips
big nose	• a photo of a person with a big nose • a photo of a person with a large, prominent nose • a photo of a person with a wide nose	• a photo of a person with a small nose • a photo of a person with a petite nose • a photo of a person with a narrow, delicate nose
black hair	• a photo of a person with black hair • a photo of a person with dark black hair • a photo of a person with jet-black hair	• a photo of a person without black hair • a photo of a person with non-black hair • a photo of a person with blonde, brown, or gray hair
blond hair	• a photo of a person with blond hair • a photo of a person with blonde hair • a photo of a person with light golden hair • a photo of a person with fair-colored hair	• a photo of a person without blond hair • a photo of a person with dark or brown hair • a photo of a person with non-blond hair

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FROM CONCEPT IDENTIFICATION TO CONCEPT REMOVAL

Concept	Positive prompts	Negative Prompts
blurry	• a blurry photo of a person • an out-of-focus photo of a person • a low-quality blurry image of a person	• a sharp, clear photo of a person • a high-quality, in-focus photo of a person • a crisp and well-focused photo of a person
brown hair	• a photo of a person with brown hair • a photo of a person with dark brown hair • a photo of a person with chestnut-colored hair	• a photo of a person without brown hair • a photo of a person with blonde, black, or gray hair • a photo of a person with non-brown hair
bushy eyebrows	• a photo of a person with bushy eyebrows • a photo of a person with thick, heavy eyebrows • a photo of a person with dense, prominent brows	• a photo of a person with thin eyebrows • a photo of a person with light, sparse eyebrows • a photo of a person with narrow, delicate brows
chubby	• a photo of a chubby person • a photo of a person with a round, full face • a photo of an overweight person • a photo of a heavyset person with chubby cheeks	• a photo of a slim person • a photo of a person with a lean face • a photo of a person with a thin face and defined cheekbones
double chin	• a photo of a person with a double chin • a photo of a person with extra flesh under the chin • a photo of a person with a prominent double chin	• a photo of a person without a double chin • a photo of a person with a well-defined jawline • a photo of a person with no extra chin
eyeglasses	• a photo of a person wearing eyeglasses • a photo of a person with glasses • a photo of a person wearing spectacles • a photo of a person in glasses	• a photo of a person not wearing eyeglasses • a photo of a person without glasses • a photo of a person with no spectacles
goatee	• a photo of a person with a goatee • a photo of a man with a goatee beard • a photo of a person with a small beard on the chin	• a photo of a person without a goatee • a photo of a person with no chin beard • a photo of a clean-shaven person with no goatee
gray hair	• a photo of a person with gray hair • a photo of a person with grey hair • a photo of a person with silver hair • a photo of an older person with gray hair	• a photo of a person without gray hair • a photo of a person with naturally colored hair • a photo of a person with dark or blonde hair, not gray
heavy makeup	• a photo of a person wearing heavy makeup • a photo of a person with a lot of makeup on • a photo of a person with dramatic, full-face makeup • a photo of a person wearing bold and heavy cosmetics	• a photo of a person wearing no makeup • a photo of a person with a natural, bare face • a photo of a person wearing minimal or no makeup
high cheekbones	• a photo of a person with high cheekbones • a photo of a person with prominent, elevated cheekbones • a photo of a person with defined, high facial bone structure	• a photo of a person with low or flat cheekbones • a photo of a person with a round face and no prominent cheekbones • a photo of a person without high cheekbones
male	• a photo of a man • a photo of a male person • a photo of a guy	• a photo of a woman • a photo of a female person • a photo of a lady
mouth slightly open	• a photo of a person with their mouth slightly open • a photo of a person with lips parted • a photo of a person with a slightly open mouth	• a photo of a person with their mouth closed • a photo of a person with lips pressed together • a photo of a person with a closed mouth
mustache	• a photo of a person with a mustache • a photo of a man with a mustache above the lip • a photo of a person with facial hair on their upper lip	• a photo of a person without a mustache • a photo of a person with a clean upper lip • a photo of a person with no mustache

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Concept	Positive prompts	Negative Prompts
narrow eyes	• a photo of a person with narrow eyes • a photo of a person with small, squinting eyes • a photo of a person with slender, narrow-set eyes	• a photo of a person with wide, open eyes • a photo of a person with large, round eyes • a photo of a person with wide-open eyes
no beard	• a photo of a person with no beard • a photo of a clean-shaven person • a photo of a person with no facial hair • a photo of a person with a smooth, beardless face	• a photo of a person with a beard • a photo of a person with facial hair • a photo of a bearded person
oval face	• a photo of a person with an oval face shape • a photo of a person with an elongated, oval-shaped face • a photo of a person with a classic oval face	• a photo of a person with a round or square face shape • a photo of a person with a non-oval face • a photo of a person with a wide or angular face
pale skin	• a photo of a person with pale skin • a photo of a person with fair, light complexion • a photo of a person with very light skin tone	• a photo of a person with dark or tanned skin • a photo of a person with a warm or deep skin tone • a photo of a person with non-pale skin
pointy nose	• a photo of a person with a pointy nose • a photo of a person with a sharp, pointed nose • a photo of a person with a narrow, pointed nose tip	• a photo of a person with a round or flat nose • a photo of a person with a broad, non-pointy nose • a photo of a person with a blunt nose tip
receding hairline	• a photo of a person with a receding hairline • a photo of a person whose hair is thinning at the temples • a photo of a person with a high, receding hairline	• a photo of a person with a full, non-receding hairline • a photo of a person with thick hair at the forehead • a photo of a person with no receding hairline
rosy cheeks	• a photo of a person with rosy cheeks • a photo of a person with pink, flushed cheeks • a photo of a person with a rosy, healthy complexion on the cheeks	• a photo of a person with pale or neutral cheeks • a photo of a person with no rosy color on the cheeks • a photo of a person with a non-flushed complexion
sideburns	• a photo of a person with sideburns • a photo of a man with sideburns along the face • a photo of a person with hair growing on the sides of the face	• a photo of a person without sideburns • a photo of a person with clean-shaved sides of the face • a photo of a person with no sideburns
smiling	• a photo of a smiling person • a photo of a person with a smile on their face • a photo of a happy, smiling person • a photo of a person grinning	• a photo of a person with a neutral expression • a photo of a person who is not smiling • a photo of a serious-looking person
straight hair	• a photo of a person with straight hair • a photo of a person with sleek, straight hair • a photo of a person with flat, non-curly hair	• a photo of a person with curly or wavy hair • a photo of a person with non-straight hair • a photo of a person with wavy or curled hair
wavy hair	• a photo of a person with wavy hair • a photo of a person with gently curled, wavy hair • a photo of a person with flowing, wavy hair	• a photo of a person with straight or tightly curled hair • a photo of a person with non-wavy hair • a photo of a person with flat, straight hair
wearing earrings	• a photo of a person wearing earrings • a photo of a person with earrings in their ears • a photo of a person accessorized with earrings	• a photo of a person not wearing earrings • a photo of a person with no earrings • a photo of a person with bare ears
wearing hat	• a photo of a person wearing a hat • a photo of a person with a hat on their head • a photo of a person in a hat	• a photo of a person not wearing a hat • a photo of a person with no hat • a photo of a person with their bare head visible

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Concept	Positive prompts	Negative Prompts
wearing lipstick	• a photo of a person wearing lipstick • a photo of a person with lipstick on their lips • a photo of a person with colored lips from lipstick	• a photo of a person not wearing lipstick • a photo of a person with natural, bare lips • a photo of a person with no lipstick
wearing necklace	• a photo of a person wearing a necklace • a photo of a person with a necklace around their neck • a photo of a person accessorized with a necklace	• a photo of a person not wearing a necklace • a photo of a person with a bare neck and no necklace • a photo of a person with no necklace
wearing necktie	• a photo of a person wearing a necktie • a photo of a person with a tie around their neck • a photo of a formally dressed person wearing a necktie	• a photo of a person not wearing a necktie • a photo of a person with no tie • a photo of a casually dressed person without a necktie
young	• a photo of a young person • a photo of a youthful-looking person • a photo of a person who looks young • a photo of a person in their twenties or thirties	• a photo of an older person • a photo of a middle-aged or elderly person • a photo of a person who looks old

Table 3: Summary of all debiasing experiments. The *Debiasing Method* column encodes the applied debiasing procedure and its configuration (see Section 2 **Methods**). Columns are grouped according to the target concept being removed, while the corresponding sub-columns report two evaluation metrics: Balanced Accuracy (BA) and Relative Concept Detection Rate (RCDR; Eq. 1). BA is measured using a zero-shot concept detection protocol based on prompt embeddings listed in Table 2, computed with the CLIP ViT-L/14 model. Evaluation is performed for each CelebA attribute on a balanced subset of 256 images consisting of 128 positive and 128 negative examples. The results reveal a substantial trade-off between predictive performance and concept suppression across debiasing methods and target concepts. Procedures that achieve superior performance with respect to one metric frequently underperform with respect to the other, highlighting the inherently multi-objective nature of the debiasing problem. A complementary Pareto-front analysis of this trade-off is presented in Figure 1.

Debiased Concept Debiasing Method	bald		chubby		eyeglasses		male		wearing_hat		Median	
	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR
iterative_sequential at 12, lr	0.619	0.049	0.601	0.050	0.607	-0.070	0.616	0.000	0.627	0.010	0.616	0.010
iterative_sequential at 12, pclarc	0.619	0.016	0.601	0.050	0.607	-0.050	0.617	0.000	0.629	0.000	0.617	0.000
iterative_single at 00, lr	0.612	0.213	0.609	0.025	0.602	-0.030	0.615	0.000	0.618	0.010	0.612	0.010
iterative_single at 00, pclarc	0.616	-0.016	0.611	-0.025	0.607	0.000	0.614	0.000	0.618	0.010	0.614	0.000
iterative_single at 01, lr	0.616	0.016	0.611	-0.025	0.606	0.010	0.614	0.000	0.618	0.010	0.614	0.010
iterative_single at 01, pclarc	0.614	0.016	0.612	-0.025	0.606	0.010	0.613	0.000	0.619	0.021	0.613	0.010
iterative_single at 02, lr	0.616	0.016	0.611	0.000	0.608	0.020	0.614	0.000	0.619	0.000	0.614	0.000
iterative_single at 02, pclarc	0.615	0.016	0.610	-0.025	0.608	0.010	0.615	0.000	<u>0.620</u>	0.010	0.615	0.010
iterative_single at 03, lr	0.615	0.016	0.609	-0.025	0.608	0.010	0.613	0.000	0.618	0.000	0.613	0.000
iterative_single at 03, pclarc	0.616	0.016	0.609	0.000	0.608	0.000	0.613	0.000	0.621	0.010	0.613	0.000
iterative_single at 04, lr	0.614	0.000	0.609	-0.025	0.608	-0.010	0.613	0.000	0.621	0.000	0.613	0.000
iterative_single at 04, pclarc	0.614	0.000	0.609	0.000	0.609	0.010	0.613	0.000	0.619	0.000	0.613	0.000
iterative_single at 05, lr	0.610	0.016	0.608	-0.025	0.605	0.000	0.611	0.000	0.619	0.000	0.610	0.000
iterative_single at 05, pclarc	0.610	0.016	0.609	0.000	0.606	0.010	0.610	0.000	0.620	-0.010	0.610	0.000
iterative_single at 06, lr	0.608	0.033	0.609	-0.050	0.603	0.010	0.611	0.000	0.620	0.010	0.609	0.010
iterative_single at 06, pclarc	0.610	0.033	0.609	-0.075	0.604	0.020	0.610	0.000	0.617	0.000	0.610	0.000
iterative_single at 07, lr	0.609	0.049	0.608	-0.075	0.605	0.000	0.609	0.000	0.619	0.000	0.609	0.000
iterative_single at 07, pclarc	0.612	0.016	0.607	-0.050	0.604	0.030	0.609	0.000	0.615	-0.010	0.609	0.000
iterative_single at 08, lr	0.610	-0.016	0.608	-0.050	0.603	-0.020	0.609	0.000	0.618	-0.031	0.609	-0.020
iterative_single at 08, pclarc	0.608	0.000	0.610	-0.050	0.604	0.000	0.613	0.000	0.615	-0.010	0.610	0.000
iterative_single at 09, lr	0.612	-0.016	0.604	-0.025	0.603	-0.010	0.612	0.000	0.616	0.010	0.612	-0.010
iterative_single at 09, pclarc	0.613	-0.016	0.604	-0.025	0.606	0.000	0.613	0.000	0.617	0.010	0.613	0.000
iterative_single at 10, lr	0.614	0.000	0.605	0.025	0.606	-0.010	0.611	0.000	0.622	0.031	0.611	0.000
iterative_single at 10, pclarc	0.617	-0.016	0.604	0.025	0.605	-0.010	0.611	0.000	0.621	0.031	0.611	0.000

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Table 3 – continued from previous page

Debiased concept	bald		chubby		eyeglasses		male		wearing_hat		Median	
	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR
iterative_single at 11, lr	0.617	0.049	0.602	0.025	0.604	-0.060	0.616	0.000	0.626	0.010	0.616	0.010
iterative_single at 11, pclarc	0.619	0.033	0.601	0.025	0.604	-0.050	<u>0.618</u>	0.000	0.623	0.010	0.618	0.010
iterative_single at 12, lr	0.619	0.033	0.602	0.050	0.606	-0.040	<u>0.616</u>	0.000	0.627	-0.010	0.616	0.000
iterative_single at 12, pclarc	0.618	0.016	0.602	0.050	0.607	-0.040	0.616	0.000	0.628	0.000	0.616	0.000
iterative_single at 13, lr	0.619	0.016	0.603	0.050	0.606	-0.060	0.614	0.000	<u>0.628</u>	0.000	0.614	0.000
iterative_single at 13, pclarc	0.619	0.016	0.603	0.050	0.607	-0.070	0.616	0.000	0.627	0.000	0.616	0.000
iterative_single at 14, lr	0.619	0.033	0.606	0.050	0.610	-0.050	0.616	0.000	0.626	0.000	0.616	0.000
iterative_single at 14, pclarc	0.618	0.016	0.606	0.000	0.609	-0.050	0.618	0.000	0.625	0.000	<u>0.618</u>	0.000
iterative_single at 15, lr	0.618	0.016	0.602	0.100	0.603	-0.060	0.615	0.000	0.622	-0.042	0.615	0.000
iterative_single at 15, pclarc	0.617	0.016	0.600	0.075	0.605	-0.050	0.614	0.000	0.625	-0.042	0.614	0.000
iterative_single at 16, lr	0.618	0.033	0.597	0.050	0.605	-0.010	0.613	0.000	0.622	-0.042	0.613	0.000
iterative_single at 16, pclarc	<u>0.619</u>	0.033	0.599	0.050	0.607	0.000	0.614	0.000	0.621	-0.052	0.614	0.000
iterative_single at 17, lr	0.611	0.049	0.596	0.025	0.601	-0.020	0.610	0.000	0.616	-0.042	0.610	0.000
iterative_single at 17, pclarc	0.612	0.033	0.596	0.000	0.604	-0.040	0.608	0.000	0.621	-0.031	0.608	0.000
iterative_single at 18, lr	0.608	0.016	0.602	0.000	0.604	-0.050	0.611	0.000	0.617	-0.062	0.608	0.000
iterative_single at 18, pclarc	0.607	0.000	0.604	0.000	0.604	-0.050	0.613	0.000	0.619	-0.052	0.607	0.000
iterative_single at 19, lr	0.609	0.049	0.596	-0.025	0.594	-0.020	0.609	0.000	0.610	-0.062	0.609	-0.020
iterative_single at 19, pclarc	0.606	0.082	0.599	0.075	0.594	-0.050	0.613	0.000	0.612	-0.021	0.606	0.000
iterative_single at 20, lr	0.608	0.033	0.597	-0.100	0.595	-0.020	0.606	0.000	0.606	-0.062	0.606	-0.020
iterative_single at 20, pclarc	0.609	0.115	0.603	0.000	0.591	-0.060	0.609	0.000	0.604	0.000	0.604	0.000
iterative_single at 21, lr	0.589	0.049	0.592	-0.125	0.594	-0.020	0.597	0.000	0.600	-0.094	0.594	-0.020
iterative_single at 21, pclarc	0.596	0.148	0.592	-0.075	0.593	-0.060	0.599	0.032	0.596	-0.010	0.596	-0.010
iterative_single at 22, lr	0.588	0.197	0.583	-0.150	0.590	0.150	0.591	0.016	0.590	-0.052	0.590	0.016
iterative_single at 22, pclarc	0.593	0.164	0.588	-0.075	0.587	0.150	0.584	0.224	0.592	0.146	0.588	0.150
iterative_single at 23, lr	0.474	1.098	0.509	0.900	0.499	0.950	0.494	0.912	0.506	<u>0.927</u>	0.499	0.927
iterative_single at 23, pclarc	0.480	<u>1.033</u>	0.509	0.375	0.493	0.660	0.496	0.984	0.508	1.062	0.496	0.984
sequential, diff_means	0.606	0.934	0.590	0.475	0.605	0.910	0.595	0.912	0.614	0.698	0.605	0.910
sequential, lr	0.607	0.820	0.604	0.250	0.607	0.870	0.606	0.800	0.616	0.729	0.607	0.800
sequential, pclarc	0.610	0.951	0.598	0.375	0.609	<u>0.940</u>	0.591	1.000	0.609	0.604	0.609	<u>0.940</u>
single at 00, diff_means	0.614	0.000	0.611	-0.025	0.607	-0.020	0.615	0.000	0.618	0.000	0.614	0.000
single at 00, lr	0.616	0.033	0.611	-0.025	0.607	-0.010	0.614	0.000	0.619	0.000	0.614	0.000
single at 00, pclarc	0.615	0.000	0.613	-0.025	0.607	-0.010	0.615	0.000	0.618	0.010	0.615	0.000
single at 01, diff_means	0.617	0.000	0.608	-0.025	0.607	-0.010	0.614	0.000	0.617	0.000	0.614	0.000
single at 01, lr	0.617	0.000	0.612	-0.025	0.608	-0.010	0.614	0.000	0.618	-0.010	0.614	-0.010
single at 01, pclarc	0.617	-0.016	0.611	-0.025	0.607	0.000	0.616	0.000	0.618	0.010	0.616	0.000
single at 02, diff_means	0.616	0.000	0.611	0.000	0.608	-0.010	0.612	0.000	0.621	-0.021	0.612	0.000
single at 02, lr	0.615	0.000	0.612	0.000	0.607	0.000	0.615	0.000	0.621	-0.021	0.615	0.000
single at 02, pclarc	0.617	0.000	0.612	-0.025	0.607	0.000	0.615	0.000	0.619	0.010	0.615	0.000
single at 03, diff_means	0.616	0.016	0.610	0.000	0.607	0.000	0.613	0.000	0.618	0.010	0.613	0.000
single at 03, lr	0.615	0.033	0.611	-0.025	0.608	0.000	0.614	0.000	0.618	0.010	0.614	0.000
single at 03, pclarc	0.615	0.000	0.611	-0.025	0.608	0.000	0.613	0.000	0.618	0.000	0.613	0.000
single at 04, diff_means	0.617	0.016	0.610	-0.025	0.606	0.000	0.614	0.000	0.619	0.021	0.614	0.000
single at 04, lr	0.614	0.033	0.612	-0.025	0.606	0.000	0.615	0.000	0.619	0.010	0.614	0.000
single at 04, pclarc	0.617	0.016	0.611	0.000	0.607	-0.010	0.614	0.000	0.617	0.000	0.614	0.000
single at 05, diff_means	0.615	0.000	0.609	-0.025	0.607	-0.010	0.614	0.000	0.622	0.021	0.614	0.000
single at 05, lr	0.614	0.033	<u>0.613</u>	0.000	0.607	-0.010	0.612	0.000	0.620	0.000	0.613	0.000
single at 05, pclarc	0.615	0.016	0.609	0.000	0.607	-0.010	0.614	0.000	0.617	0.000	0.614	0.000
single at 06, diff_means	0.614	0.016	0.610	-0.050	0.606	0.000	0.613	0.000	0.621	0.042	0.613	0.000
single at 06, lr	0.614	0.016	0.612	0.000	0.607	-0.020	0.610	0.000	0.620	0.010	0.612	0.000
single at 06, pclarc	0.617	0.000	0.609	-0.025	0.606	-0.020	0.615	0.000	0.616	0.000	0.615	0.000
single at 07, diff_means	0.615	0.033	0.609	-0.025	0.606	-0.020	0.614	0.000	0.622	0.042	0.614	0.000
single at 07, lr	0.615	0.016	0.609	-0.025	0.607	-0.020	0.612	0.000	0.620	0.000	0.612	0.000
single at 07, pclarc	0.617	0.016	0.608	0.000	0.607	-0.020	0.615	0.000	0.617	-0.010	0.615	0.000
single at 08, diff_means	0.615	0.049	0.609	-0.025	0.607	0.000	0.614	0.000	0.621	0.010	0.614	0.000
single at 08, lr	0.615	0.033	0.610	0.000	0.608	-0.020	0.612	0.000	0.620	0.000	0.612	0.000
single at 08, pclarc	0.616	0.016	0.608	-0.025	0.604	-0.010	0.614	0.000	0.616	-0.010	0.614	-0.010
single at 09, diff_means	0.616	0.016	0.607	-0.025	0.606	-0.030	0.615	0.000	0.619	0.010	0.615	0.000
single at 09, lr	0.616	0.016	0.608	-0.025	0.609	-0.030	0.614	0.000	0.620	0.010	0.614	0.000
single at 09, pclarc	0.615	0.000	0.609	0.000	0.608	-0.010	0.614	0.000	0.618	-0.010	0.614	0.000
single at 10, diff_means	0.616	0.016	0.609	-0.025	0.604	-0.010	0.614	0.000	0.616	-0.010	0.614	-0.010
single at 10, lr	0.615	0.016	0.609	0.000	0.609	-0.010	0.615	0.000	0.617	0.000	0.615	0.000
single at 10, pclarc	0.615	0.000	0.609	0.000	0.606	-0.010	0.614	0.000	0.616	-0.010	0.614	0.000
single at 11, diff_means	0.615	0.016	0.608	-0.025	0.604	-0.030	0.616	0.000	0.616	-0.010	0.615	-0.010
single at 11, lr	0.615	0.033	0.611	0.000	0.610	-0.010	0.615	0.000	0.618	0.000	0.615	0.000
single at 11, pclarc	0.616	0.033	0.608	0.000	0.603	-0.010	0.616	0.000	0.616	-0.010	0.616	0.000
single at 12, diff_means	0.618	-0.016	0.608	-0.025	0.605	-0.010	0.616	0.000	0.619	-0.010	0.616	-0.010
single at 12, lr	0.615	0.016	0.610	-0.025	0.608	0.010	0.615	0.000	0.617	0.000	0.615	0.000
single at 12, pclarc	0.616	0.000	0.609	0.000	0.606	-0.010	0.617	0.000	0.617	-0.010	0.616	0.000
single at 13, diff_means	0.617	-0.016	0.609	-0.025	0.606	-0.010	0.617	0.000	0.618	0.000	0.617	-0.010
single at 13, lr	0.617	0.000	0.612	0.000	0.608	0.010	0.615	0.000	0.618	0.000	0.615	0.000
single at 13, pclarc	0.617	0.000	0.610	0.000	0.607	-0.020	0.616	0.000	0.618	0.000	0.616	0.000
single at 14, diff_means	0.616	0.000	0.610	0.000	0.607	0.000	0.617	0.000	0.618	0.000	0.616	0.000
single at 14, lr	0.617	-0.033	0.609	-0.025	0.608	0.000	0.615	0.000	0.618	0.000	0.615	0.000
single at 14, pclarc	0.616	0.000	0.610	0.000	0.606	-0.010	0.618	0.000	0.617	0.000	0.616	0.000
single at 15, diff_means	0.616	0.016	0.611	-0.025	0.606	-0.010	0.616	0.000	0.620	0.010	0.616	0.000
single at 15, lr	0.616	-0.016	0.610	-0.025	0.607	-0.010	0.615	0.000	0.618	0.010	0.615	-0.010
single at 15, pclarc	0.616	0.000	0.610	-0.025	0.607	0.000	0.616	0.000	0.618	0.000	0.616	0.000
single at 16, diff_means	0.616	-0.016	0.609	0.000	0.607	0.000	0.614	0.000	0.619	0.000	0.614	0.000
single at 16, lr	0.616	0.033	0.609	0.000	0.610	0.000	0.614	0.000	0.619	0.000	0.614	0.000
single at 16, pclarc	0.617	-0.033	0.610	0.000	0.607	0.000	0.614	0.000	0.618	0.000	0.614	0.000
single at 17, diff_means	0.614	-0.016	0.608	0.000	0.607	-0.020	0.613	0.000	0.623	0.010	0.613	0.000

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FROM CONCEPT IDENTIFICATION TO CONCEPT REMOVAL

Table 3 – continued from previous page

Debiased concept	bald		chubby		eyeglasses		male		wearing_hat		Median	
	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR	BA	RCDR
single at 17, lr	0.615	0.033	0.610	0.025	0.609	-0.010	0.614	0.000	0.618	0.010	0.614	0.010
single at 17, pclarc	0.615	-0.016	0.608	0.025	0.607	0.000	0.614	0.000	0.619	0.010	0.614	0.000
single at 18, diff_means	0.613	0.016	0.607	0.050	0.607	-0.010	0.613	0.000	0.623	0.000	0.613	0.000
single at 18, lr	0.617	0.033	0.611	-0.025	0.610	-0.010	0.614	0.000	0.619	0.000	0.614	0.000
single at 18, pclarc	0.614	0.000	0.609	0.050	0.606	-0.010	0.613	0.000	0.622	0.010	0.613	0.000
single at 19, diff_means	0.612	0.098	0.610	0.050	0.608	0.050	0.611	0.000	0.621	0.062	0.611	0.050
single at 19, lr	0.616	0.049	0.610	0.000	0.608	0.020	0.614	0.000	0.618	0.052	0.614	0.020
single at 19, pclarc	0.612	0.098	0.610	0.025	0.608	0.030	0.613	0.000	0.619	0.083	0.612	0.030
single at 20, diff_means	0.613	0.115	0.607	0.100	0.607	0.100	0.610	0.000	0.621	0.083	0.610	0.100
single at 20, lr	0.615	0.049	0.610	0.025	0.609	0.010	0.613	0.000	0.617	0.094	0.613	0.025
single at 20, pclarc	0.614	0.115	0.608	0.075	0.607	0.080	0.613	0.000	0.616	0.104	0.613	0.080
single at 21, diff_means	0.612	0.213	0.607	0.150	0.610	0.200	0.608	0.000	0.620	0.156	0.610	0.156
single at 21, lr	0.615	0.066	0.609	0.050	0.608	0.100	0.612	0.000	0.619	0.135	0.612	0.066
single at 21, pclarc	0.610	0.148	0.609	0.125	0.611	0.160	0.610	0.040	0.615	0.125	0.610	0.125
single at 22, diff_means	0.613	0.393	0.601	0.225	0.613	0.250	0.610	0.056	0.620	0.323	0.613	0.250
single at 22, lr	0.613	0.279	0.608	0.025	0.609	0.320	0.614	0.008	0.617	0.167	0.613	0.167
single at 22, pclarc	0.612	0.393	0.607	0.200	<u>0.612</u>	0.290	0.609	0.304	0.613	0.219	0.612	0.290
single at 23, diff_means	0.606	0.951	0.593	<u>0.500</u>	0.606	0.900	0.598	0.968	0.617	0.906	0.606	0.906
single at 23, lr	0.610	0.639	0.609	0.025	0.612	0.710	0.614	0.760	0.618	0.688	0.612	0.688
single at 23, pclarc	0.602	0.967	0.596	0.425	0.607	0.910	0.590	<u>0.992</u>	0.613	0.562	0.602	0.910